THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF PHYSICAL SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES

First Semester 2025/2026 Academic Session

MTS 101: INTRODUCTORY MATHEMATICS I ASSIGNMENT

INSTRUCTION: Answer all the questions. **Time Allowed:** Four Weeks (672 Hours)
To be submitted on or before 12:00noon on Wednesday 25th February, 2026.

1. Mr. Iwe and Mr. Olodo had a hot argument on whether or not 2^{2n+1} is divisible by 8 for all positive integers. Mr. Iwe is of the opinion that since 2^{2n+1} is divisible by 8 when $n = 1$, then it is divisible by 8 for all positive integers whereas Mr. Olodo believes that $f(k+1) = 8(9M+1)$ is the sufficient condition that guarantees that 2^{2n+1} is divisible by 8 for all positive integers M.
   (a) Which of these two MTS 101 Students do you agree with?
   (b) Justify your answer.

2. Given that $2^n - 1$ is a multiple of 1, $3^n - 1$ is a multiple of 2, $4^n - 1$ is a multiple of 3,..., use the principle of mathematical induction to prove that $k^n - 1$ is a multiple of $(k-1)$ for all positive integers k and n.

3. In the Department of Mathematical Sciences, FUTA, reliability analysis of multi-component systems uses probability inequalities. Let $0 \leq p_i \leq 1$ for $i = 1, 2, \ldots, n$. Prove by induction that $\prod_{i=1}^{n}(1-p_i) \geq 1 - \sum_{i=1}^{n} p_i$, and determine when equality holds.

4. Dr. Afolabi's Student, Iwe Ologbon was asked to prove that $n^4 + 4n^2 + 11$ is a multiple of 16 for all positive integers. What is the nth term of the possible values of n?

5. An SS3 student of The Federal University of Technology, Akure, Staff School, Dauda Itooknow, was asked to use the principle of mathematical induction to prove that $\sum_{i=0}^{i=n} = \frac{1-x^{n+1}}{5-x}$. For what values of x can she prove this?

6. As each of the students of MTS 101 arrives for an examination, each have a short discussion with all the other students present. Use mathematical induction to show that if n students come for the examination, then $\frac{n(n-1)}{2}$ discussions occur.

7. What is a number? State and explain ten (10) properties of real numbers that you know.

8. Define the term the Principle of Mathematical Induction and highlight the steps involved in the Principle of Mathematical Induction.

9. With the aid of a well labelled Venn diagram, explain the relationship that exists among all the types of numbers below:

   (a) Natural/Counting Numbers
   (b) Integers
   (c) Rational Numbers
   (d) Irrational Numbers
   (e) Real Numbers
   (f) Complex Numbers
   (g) Even Numbers
   (h) Odd Numbers
   (i) Prime Numbers
   (j) Composite Numbers

10. Prove that $n! > 3^n$ for $n \geq 7$.

11. Prove that $x^n - y^n$ is divisible by $x - y$, where x, y are integers with $x \neq y$.

12. Use the principle of Mathematical Induction to show that

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots + \frac{1}{2^n} = \frac{2^n - 1}{2^n}$$

13. Use the principle of Mathematical Induction to show that

$$\frac{1}{4 \times 1^2 - 1} + \frac{1}{4 \times 2^2 - 1} + \ldots + \frac{1}{4 \times n^2 - 1} = \frac{1}{2n + 1}$$

14. Prove by Mathematical Induction that

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \ldots + n \times 2^{n-1} = 1 + (n-1)2^n$$

15. Use Mathematical Induction to show that

$$\sin\theta + \sin 2\theta + \sin 3\theta + \ldots + \sin n\theta = \frac{\sin\frac{1}{2}(n+1)\theta \sin\frac{1}{2}n\theta}{\sin\frac{1}{2}\theta}$$

16. Verify that for all $n \geq 1$, the sum of the squares of the first $2n$ positive integers is given by the formula

$$1^2 + 2^2 + 3^2 + \ldots + (2n)^2 = \frac{n(2n+1)(4n+1)}{3}$$

17. Find the first positive integer n that causes the statement $n^2 - 3n < 100$ to fail using the principle of Mathematical Induction.

18. Prove that $(3+\sqrt{5})^n + (3-\sqrt{5})^n$ is an even integer for all natural numbers n using the principle of Mathematical Induction.

19. Use the principle of Mathematical Induction to prove that $2002^{n+2} + 2003^{2n+1}$ is divisible by 4005.

20. Prove by Mathematical Induction that
$$\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$$

21. Prove that the sum of the cubes of any three consecutive natural numbers is divisible by 9 using mathematical induction.

22. Prove $7^n - 2^n$ is divisible by 5 without any remainder for all positive integers.

23. Prove by Mathematical Induction that
$$1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n}{6}(n+1)(2n+1)$$

24. Use the principle of Mathematical Induction to prove that for positive integers $a_1, a_2, \ldots, a_n$,
$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \ldots a_n}$$

25. Show that $3^{4n+2} + 2 \cdot 4^{3n+1}$ is exactly divisible by 17 if n is a positive integer.

26. Use the Principle of Mathematical Induction to prove that $5^n + 2 \times 11^n$ is a multiple of 3 for all positive integer n.

27. Prove by the Principle of Induction
$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

28. Prove or disprove that $n^2 + 21n + 1$ is a prime number for all positive values of n.

29. Prove that for any positive integer $n > 7$ can be written as the sum of three or fewer squares of positive integers.

30. Prove that for any natural number n,
$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n} \leq \frac{1}{\sqrt{3n+1}}$$

DR. A. S. AFOLABI,
DEPARTMENT OF MATHEMATICAL SCIENCES,
THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE.